Genotype by environment interaction and genetic heterogeneity of environmental variance of body weight at harvest in Genetically Improved Farmed Tilapia (GIFT) (Oreochromis nilotic...

Conference Paper · July 2016

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Genotype by environment interaction and genetic heterogeneity of environmental variance of body weight at harvest in Genetically Improved Farmed Tilapia (GIFT) (*Oreochromis niloticus)* reared in three different countries

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Establishing breeding programs that improve farmed fish performance across multiple environments is crucial in a globalized aquaculture market. The main objectives of the current study were (i) to assess the growth performance of Genetically Improved Farmed Tilapia (GIFT) under different environments, across 3 different countries (Malaysia, India and China), termed genotype by environment interaction (G x E) or macro-environmental sensitivity and (ii) to quantify the genetic heterogeneity of environmental variance for body weight at harvest (BW) in GIFT as a measure of micro-environmental sensitivity or robustness. The data consist of 94,351 individuals representing 1568 full-sibs families. Selection for BW was carried out over 13 generations in Malaysia. Subsets of the families from Malaysia were sent to India and China and reared for 4 and 3 generations, respectively. First, a multi-trait animal model was used to analyze the BW in different countries as different traits. Genetic correlations (r_g) between BW in Malaysia and in India and China were 0.25 and 0.48, respectively, while r_g between BW in India and China was 0.13. These low to moderate genetic correlations between BW in different environments indicate a strong G x E interaction and re-ranking of genotypes across environments in regards to the genetic merit for growth. Second, a genetically structured environmental variance model, implemented using Bayesian inference, was used to analyze micro-environmental sensitivity of BW in each country. The results revealed the presence of genetic heterogeneity in both the BW and its environmental variance in all environments. The posterior mean of the additive genetic correlation between the BW and its environmental variance in Malaysia and China (95% highest posterior interval) were -0.53 (-0.47,-0.59) and -0.66 (-0.80,-0.60), respectively. High negative genetic correlations indicate associations between genes controlling the mean and those affecting its environmental variance. Thus, selection for higher BW would be expected to decrease environmental variation. In contrast, a weak association between additive genes affecting the mean and the environmental variance was found for BW in India with a posterior mean of the additive genetic correlation -0.03 (-0.17, 0.11). The study indicates a strong G x E interaction in BW for GIFT strain and that the environmental variance of BW is partly genetically determined. Incorporation of macro- and micro- environmental sensitivity information in the selection index of multi-environment aquaculture breeding programs enables simultaneous improvement of weight gain and increasing homogeneity/robustness.

Keywords: Environmental sensitivity, robustness, aquaculture breeding.